

VIJAY VENKATASAMY

CIVIL & STRUCTURAL ENGINEER

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PROFILE

Civil and structural engineer with over three years of combined UK consultancy and international infrastructure experience. At Francis Bradshaw Consulting I lead structural design from concept to detailed design in reinforced concrete, steel and timber on 40+ retail, commercial, industrial and residential projects, including SER-certified schemes in Scotland. Earlier, two years with VA Tech WABAG covered EPC procurement, site supervision and design checking on major water and wastewater plants. Graduate Member of the ICE, preparing for the CEng Professional Review.

PROFESSIONAL STATUS & CERTIFICATIONS

Membership: Graduate Member, Institution of Civil Engineers (GMICE, no. 93963668), preparing for the CEng MICE Professional Review. IStructE graduate membership in progress.

Certifications: CSCS card holder; asbestos awareness; Diploma in Tekla Structures (2023); 15 PMI and University of California project-management certificates; Primavera P6 and MS Project (2023).

TECHNICAL SKILLS

Design: Reinforced concrete, structural steel, timber, composite floors, steel connections (end plates, baseplates, fin plates), remedial design, underpinning, SuDS.

Codes & standards: Eurocodes (EC2, EC3, EC5) with UK National Annexes; BS 5950 and BS 6399 series; Scottish Building Regulations and SER; CDM 2015; IS 456.

Software: MasterSeries, Tekla Tedds, Tekla Structures, STAAD.Pro, ETABS, AutoCAD, Revit, Hilti PROFIS Engineering, MATLAB, advanced MS Excel.

Practice: Assessment of existing buildings, defect reporting, external design checking, SER submissions, embodied-carbon assessment (PAS 2080), BOQ and bar-bending take-off, programme scheduling, client and contractor liaison.

PROFESSIONAL EXPERIENCE

Civil & Structural Engineer | Francis Bradshaw Consulting, Wilmslow, UK

Nov 2024 – Present

Promoted from Graduate Structural Engineer in 2026.

- Carry out structural design from concept to detailed design in RC, steel and timber across 40+ retail, commercial, industrial and residential projects, to CDM 2015 Regulation 9 designer duties.
- Lead designer on the SER-certified Fort Kinnauld retail refurbishment in Edinburgh: glass canopy, end-bay portalisation replacing removed shopfront bracing, an 18 m hung bulkhead, M&E supports, roller-shutter posts and wall-lining anchors, to BS 5950 and EC3.
- Staged the SER certification on that project so external works could start while internal design continued, saving about two weeks on the construction programme.
- Designed a steel portal-frame maintenance building and reinforced concrete inspection pit for a local-authority waste depot, and authored the structural, drainage, pavement and earthworks specifications.
- Sized and detailed a fire-rated buffer wall about 17 m high, built into an existing steel frame in 6 m bays, with deflection heads to take vertical movement of the frame.
- Assess and alter existing structures: on a brick retail unit with concealed steelwork, kept the heavily loaded piers and replaced one intermediate pier with a slender steel column to extend the sales floor.
- Designed an overhead crane hoist over an existing bakery mixer platform from a combined 3D model, and RC crane-base foundations with a strap beam for an eccentric 6.5 t crane column at a glass works.
- Check external calculation packages and flag issues beyond the brief: compared two competing portal-frame schemes on an independent model, raising queries on a baseplate clash, a bolt-count discrepancy, torsion and design life; on a mezzanine check, caught three errors corrected before construction.
- Carry out solo site inspections to my own RAMS, including a trading store with the public present (presumed asbestos flagged from the survey) and a cracked residential property.
- Specified a cheaper, stronger steel SHS sub-frame in place of a contractor's aluminium proposal on a signage package, and cleared SER and landlord-surveyor queries through to Building Warrant.

- Advised a client against removing a load-bearing spine wall before propping, and set out the correct demolition sequence and stability frame under CDM Regulation 9.
- Built the practice's embodied-carbon tool (PAS 2080) and its first Carbon Reduction Plan, an 8.82% cut on the FY2021/22 baseline, and AI-assisted defect-reporting and fee-letter tools that an engineer signs off before issue.

Civil Engineer Trainee | VA Tech WABAG Ltd, India and Southeast Asia

Aug 2021 – Aug 2023

- Sole procurement engineer for all engineering packages on an international energy company's effluent treatment plant in Malaysia: 66 procurement items across 29 mechanical, 26 piping and 11 instrumentation packages from vendors in India, Sweden and Qatar; ran letters of credit, DPX export certification, milestone payments and bill-of-lading filing.
- Controlled package costs on the US\$11.45M EPC contract: sourced an alternative vendor and negotiated a raised price back down, and formalised a three-package contract variation through back-to-back vendor provisions. Nominated for the company's Extra Miler award.
- Supervised construction of the 178.3 MLD Pillur III water treatment plant, Coimbatore: daily safety checks, toolbox talks and permits; approved pour cards to IS 456 and refused non-compliant pours; stopped unsafe work on my own authority (work at height without a harness, an unstable scaffold platform, and a confined-space evacuation on an oxygen alarm).
- Acted as checker on the Pagla STP, Dhaka (200 MLD) and Patna STP, India (150 MLD): reviewed RC and steel calculations and drawings before issue.

Engineering Internships | AK Consultant · Shree Constructions · VA Tech WABAG, India

2018 – 2021

- Early experience (about 8 months total) in structural calculations, AutoCAD and Tekla drafting, site supervision and quality inspection on residential (G+2/G+3) and water-treatment projects.

EDUCATION

MSc Structural Engineering | University of Leeds, UK

2023 – 2024

Merit (2:1). Dissertation: Monte Carlo and subset simulation in MATLAB to propagate seismic uncertainty through to structural failure probability. Group project: RIBA Stage 1–4 structural design of a proposed HS2 station for Leeds, with CDM 2015 compliance and an embodied-carbon assessment.

BEng (Hons) Civil Engineering | Anna University (Easwari Engineering College), Chennai, India

2017 – 2021

CGPA 7.96 / 10. Dissertation: partial replacement of cement and aggregates with industrial waste in concrete.

PROFESSIONAL DEVELOPMENT

- Structured CPD maintained towards ICE and IStructE chartership, recorded against the ICE development framework across the mandatory ethics, safety and sustainability themes.
- ICE technical webinars and IStructE and manufacturer CPD on subsidence, sustainable drainage, embodied carbon and timber design.